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Segmentation of bleeding regions in wireless capsule endoscopy for detection of informative frames

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ABSTRACT

Wireless capsule endoscopy (WCE) is an effective means for diagnosis of gastrointestinal disorders. Detection of informative scenes in WCE video could reduce the length of transmitted videos and help the diagnosis procedure. In this paper, we investigate the problem of simplification of neural networks for automatic bleeding region segmentation inside capsule endoscopy device. Suitable color channels are selected as neural networks inputs, and image classification is conducted using a multi-layer perceptron (MLP) and a convolutional neural network (CNN) separately. Both CNN and MLP structures are simplified to reduce the number of computational operations. Performances of two simplified networks are evaluated on a WCE bleeding image dataset using the DICE score. Simulation results show that applying simplification methods on both MLP and CNN structures reduces the number of computational operations significantly with AUC-ROC greater than 0.97. Although CNN performs better in comparison with the simplified MLP, the simplified MLP segments bleeding regions with a significantly smaller number of computational operations. Concerning the importance of having a simple structure or a more accurate model, each of the designed structures could be selected for inside capsule implementation.

1. Introduction

Wireless capsule endoscopy (WCE) is a non-invasive and painless endoscopic method employed in screening different parts of the gastrointestinal (GI) organs which is used for a variety of medical experiments [1]. WCE imaging is always used for patients suspicious of bleeding and other types of abnormalities such as Crohn's disease in GI. Also, it is possible to use WCE for patients with polyposis syndrome and small bowel disorder [1]. Researchers have investigated automatic methods for detection of abnormalities since 2001, when WCE was first launched [1]. These methods have significantly reduced the time spent by physicians for medical diagnostic and detection [2].

In recent years, researchers have developed different methods for automatic detection of a variety of abnormalities such as ulcer, bleeding, polyps and other abnormalities in WCE images. In some studies such as [3] and [4], handcrafted features are employed. In [3], saliency is calculated based on color and texture information. For texture representation, LOG-Gabor filter, *scale invariant feature transform* (SIFT) and *local binary pattern* (LBP) are used. For color information, second channels of HSI and CMYK color spaces are used. All features are coded with K-means clustering, and final saliency map is obtained by max-pooling of the saliency map and coded features. Mehmood et al. used different-order of moments as a saliency analysis method to summarize WCE video frames [4].

Machine learning methods have gained a lot of attention in recent years due to their remarkable success in medical image processing applications. Among them, support vector machine (SVM) is used extensively for abnormality detection in different human organs. Fu et al. [5], used super pixel color features to speed up the bleeding detection process and employed SVM as a classifier. Suman et al. [6], removed noise, edges, dark, and light regions to avoid misclassification of bleeding regions. In their work, a combination of RGB channels at the block level is used as SVM input. In [7], a K-means algorithm is employed to cluster bleeding images into two clusters based on the block's statistical characteristics.

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Bleeding images are then detected from non-bleeding images using differential features, and finally, bleeding zones are localized using an SVM.

In many studies, different features such as SIFT, LBP, *speed up robust feature* (SURF), Gabor filter, the *histogram of the gradient* (HOG) and *discrete wavelet transform* (DWT) are used as SVM classifier's inputs [8–14]. In [8], LBP feature is extracted from the "*I*" channel of the HSI color space in 8×8 image blocks and then each block is segmented by SVM. Deeba et al. [9], presented a method based on cellular automata for clustering WCE images. For segmentation of bleeding regions, they utilized SVM to classify the cluster centroids. In [10], the histogram of pixels is assigned to each cluster in YCbCr color space, then it is used as a descriptor, and by using an SVM, WCE images are segmented. In [11], a method for detection of polyp and ulcer in WCE frames is presented where Gabor filter is applied for simulating the human visual system. Also, edges are detected using *smallest univalue segment assimilating nucleus* (SUSAN) edge detection method to improve the accuracy of the detection algorithm. Then, using the results from previous stages, regions containing ulcer and polyp are detected based on SVM and a decision tree. Yuan et al. [12], extracted different features around image key points including LBP, SIFT, and HOG. In their work, the set of extracted features are clustered and classified using an SVM classifier. In [13], a method for detection of abnormalities in WCE images is presented where color and texture features are extracted using moments, LBP and DWT. Importance of the features is identified using a saliency map and based on this importance, WCE images are classified by SVM. In [14], a method for automatic lesion detection in capsule endoscopy is presented. After CIE-Lab color transformation, salient points are detected via SURF detector based on the area around each pixel. Using features of salient points, SVM is employed for classification of regions to lesion and non-lesion.

Recently, deep neural networks achieved considerable successes in the detection and segmentation of abnormalities in WCE images. Jia et al. [15], proposed a bleeding frame detection method consisting of feature extraction and classification steps. In their work, histograms of K-Means clustering are used as input features and are classified by a *convolutional neural network* (CNN). Also, Jia et al. [16], augmented datasets by rotation and mirroring to extend the number of training and test objects and used a CNN structure for detection of bleeding frames. In [17], bleeding images are classified into bleeding and non-bleeding using SVM and then in each classified image, bleeding regions are segmented using CNN. In [18], image patches are fed into a single CNN structure and classified as normal and abnormal.

In WCE videos, only a few frames are informative and transmitting all the frames will increase the battery usage of WCE devices. Hence, designing a low power system for image analysis and informative frame detection is very demanding due to limited memory capacity and limited power supply [4]. Also, there is a great demand for the implementation of the diagnostic process inside the WCE device. Although in some previous works powerful tools such as CNN were utilized for segmentation, the complexity of the detection method can be problematic for inside capsule implementation. For example, the method of [16], uses CNN with 15 M parameters and method in [17], uses a fully-convolutional network with 6 M parameters [19]. A large number of parameters and the use of *multiply-accumulate* (MAC) operations make the use of CNNs infeasible considering the size and resources of the capsule endoscopy.

Several methods have been developed for hardware implementation of image processing algorithms inside the capsule device. In [20], a hardware architecture for WCE image assessment inside the capsule is presented. Informative frames are detected inside the capsule using mean, variance, skewness and kurtosis. Image compression inside WCE is introduced as another way to reduce the power consumption needed for transferring the video frames to outside. In [21], a hardware architecture for image compression inside WCE is proposed. To compress the images, an integer based DCT, and an efficient coefficient encoding with low complexity is applied. In [22], a hardware core including a camera interface, *first in first out* (FIFO) queue, controller, and image compressor is presented. In [23], an architecture for calculation of DCT transform which is employed in image compression inside WCE is designed. The applied DCT is a 1-D DCT transform which reduces the number of addition and shift operations.

In this paper, we utilize artificial neural networks for bleeding region segmentation. Since the computational complexity is important, we employ both an MLP and a CNN structure as computationally simple and complex structures, respectively. Different color spaces are analyzed to select channels containing more information about bleeding regions. For having a simple and efficient network structure, different network simplification approaches are investigated. Multiplications are removed from the MLP structure by the weight quantization approach. In the convolutional network, multiplications are removed from the fully connected layers (FCLs) and redundant computations are eliminated from convolutional layers (CLs) by using a simultaneous quantization and pruning approach. Both network structures segment bleeding regions with high DICE score. In both of the proposed network structures for bleeding region segmentation, computational complexity is reduced significantly, and this reduction provides an opportunity to implement the proposed methods inside the capsule device. The main contributions of this study are summarized as followings.

- Selection of proper color channels based on mutual information as inputs of utilized neural networks.
- Designing a simplified MLP structure suitable for image analysis inside the capsule endoscopy device.
- Employing simultaneous quantization and pruning to simplify the CNN structure with a little drop of accuracy.

The remainder of this paper is organized as follows. In Section II, neural network simplification techniques are briefly explained. In Section III, the proposed method for the segmentation of bleeding regions in WCE images is demonstrated. Section IV is dedicated to experimental results and finally concluding remarks are presented in Section V.

2. Neural network simplification techniques

Machine learning techniques are employed for medical image analysis in various research projects conducted in recent years. Among different machine learning methods, *artificial neural networks* (ANNs) have considerable achievements in automatic segmentation and classification of a variety of abnormalities [24–26]. CNNs, as an advanced version of primary neural networks, can appropriately accomplish the classification and segmentation tasks. Although CNNs are usually more accurate than simple MLP networks and have better capabilities in automatic feature selection, their structure is more complex than MLP. When it comes to implementing CNNs on devices with limited computational and power resources, their structural complexity is forbidding. Different techniques were proposed to alleviate the complexity of neural network structures. These methods are briefly explained in the following subsections.

2.1. Neural network quantization

In an ANN, computational operations are mainly required for three types of tasks including multiplication of weight matrices,

Table 1
Rough energy costs for various operations in 45 nm [29].

Operation	Multiply	Add
8-bit Integer	0.2pJ	0.03pJ
32-bit Integer	3.1pJ	0.1pJ
16-bit Floating Point	1.1pJ	0.4pJ
32-bit Floating Point	3.7pJ	0.9pJ

addition of activation function inputs, and calculation of activation function outputs. The number of computational operations can be reduced in different ways. Recently, binarization is used as an efficient way to reduce neural network's structural complexity [27–28]. During binarization, all of the weights are affected by the simplification process. In the binarization process, a value is converted to two possible values, as an example to 0 and 1. Two methods were introduced for binarization in previous studies [27–28]. The first approach is a deterministic binarization which is illustrated in the following equation:

$$\Delta W_b = \begin{cases} +1 & if\ \ W \geq 0, \\ -1 & otherwise \end{cases} \tag{1}$$

where W is a network weight before quantization and W_b is the binarized one. The second approach for binarization is the stochastic method shown in the following equation:

$$W_b = \begin{cases} +1 & with\ probability\ \ p = \sigma(w), \\ -1 & \quad with\ probability\ \ 1-p. \end{cases} \tag{2}$$

where $\sigma(x)$ is "hard sigmoid function" defined as follows:

$$\sigma\ (x) = clip\ \left(\frac{x+1}{2}, 0, 1\right) = max(0, min(1, \frac{x+1}{2})) \tag{3}$$

The binarization could decrease WCE power consumption. In [29], the power consumption of the operations with different representations was roughly presented. As shown in Table 1, the energy consumption of addition and multiplication operations is reduced significantly with a shorter bit width of the computational operations [29]. Also, concerning [29], binary representation consumes 32 times smaller memory size and 32 times fewer memory accesses in comparison with 32-bit representation.

2.2. Neural network pruning

After training neural networks, there might be some redundant weights that can be removed or merged with other weights without significant degradation in network accuracy. By removing redundant weights, the problem of redundant computations is alleviated especially for complex neural networks. Also, network pruning can be used to address the over-fitting problem while training neural networks. Different methods have been proposed for pruning such as weight decay [30], and Hessian of the loss function [31]. However, their proposed pruning techniques require second order derivatives which are not computationally efficient. In [32], a magnitude-based pruning was developed which affects weights and connections individually. In [33], a filter pruning method is presented in which all coefficients of a filter are removed. Although filter pruning results in a better structural sparsity, removing a whole filter could lead to a substantial accuracy drop. Therefore, it cannot be used as a pruning method in general. Principally, all of the pruning methods consist of three steps. In the first step network variables are trained to achieve an acceptable accuracy. In the second step, redundant weights are removed. Finally, the network is retrained to compensate for the accuracy drop caused by removing redundant parameters in the second step. In the primary pruning method, weights are removed during the training process based on a pruning rate, and finally, a pruning mask represents which weights are removed. Pruning mask is calculated using Eq. (4) in which W_l and $Mask_l$ are network weight and pruning mask matrix in layer l respectively and α is the pruning rate.

$$Mask_l\,(i,j) = \begin{cases} 0 & W_l\,(i,j) < \alpha * \sigma^2\,(W_l) \\ 1 & \qquad Otherwise \end{cases} \tag{4}$$

As shown in (4), a sample weight in layer, $W_l\,(i,j)$, is pruned if its value is less than a threshold. The threshold is equal to the pruning rate multiplied by the standard deviation of all weights in layer l (i.e. $\sigma^2(W_l)$). Pruning rate can be modified during the training process to get better results. Finally, pruning is performed during back propagation as follows:

$$W\,(i,j) = \ W\,(i,j) - \ \eta\,(\frac{\partial C}{\partial W\,(i,j)} * Mask_l(i,j)) \tag{5}$$

In (5), weights are updated by a learning rate η and gradients of the weights that were pruned are removed from the training process by multiplying the gradients matrix by $Mask_l$. Network error in terms of $W\,(i,j)$ is specified by $\frac{\partial C}{\partial W(i,j)}$, which can be compensated by retraining [32].

3. Simplified segmentation method

The proposed method includes two major parts in general. The first part is dedicated to simplification of the MLP, and the other one explains the simplification of the CNN. Prior to applying the method to a neural network, a color space conversion is required to reach better performance. The proposed method is described in the following subsections in more details.

3.1. Color space conversion

Input images can be represented in different color spaces and in each color space, some features are more representative. As an example, in retinal image analysis, the green channel is more informative than the others [25], [34]. Color space mapping can be considered as a preprocessing step applied in medical image processing techniques. HSV and CIE-*lab* color spaces are two alternatives to RGB color space. CIE-*lab* is more intuitive than RGB and is designed to approximate the human vision system [35]. "*l*," "*a*" and "*b*äre components of the CIE-*lab* so that "*l*" lies in 0 to 100 and both "*a*" and "*b*" lie in −110 to 110. HSV color space including hue, saturation, and value is quite similar to the human's perception of colors. In Fig. 1, a bleeding frame of WCE videos is shown in different color spaces. As illustrated in Fig. 1, some color spaces represent bleeding regions better than the others, and some of them have no useful information. From a visual perspective, Fig. 1 shows that the gray-scale domain, the saturation channel, and the "*a*" channel have distinctive representations of the bleeding region. Using an experiment based on information theory, it is possible to identify color spaces or channels which are suitable for representation of the bleeding regions.

3.1.1. Color space and channel selection based on mutual information

Mutual information is the amount of information that one random variable contains about another random variable and is calculated using the following equation [36].

$$I\,(X;Y) = H\,(X) + H(Y) - H(Y;X) \tag{6}$$

In (6), X and Y are random variables and H is the entropy function. The mutual information between each color channel and the ground truth can be calculated for different color spaces, as a measure of the applicability of the channels. For 55 WCE images in [37] and [38], 10 color spaces and channels including "H", 'S", and "V"

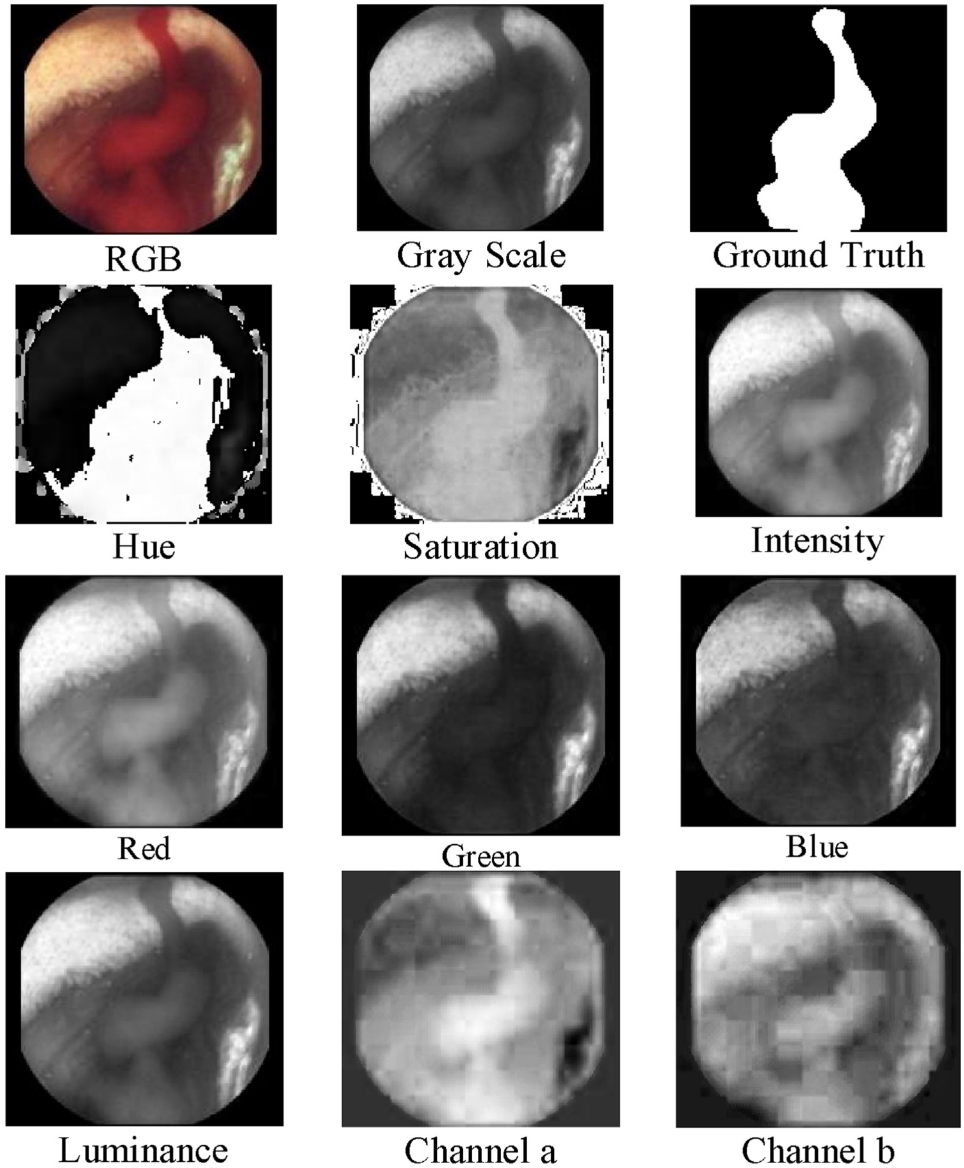

Fig. 1. Different color space in a sample WCE image.

in HSV color space, "*l*", "*a*", and "*b*" in CIE-*lab* color space, "R", "G", and "B" in RGB color space and grayscale level image are experimented. Experiments are conducted to measure the amount of information that each unary, binary, and ternary combination of color channels contain about the ground truth. In the first experiment, the mutual information of each 10 transformed images and their corresponding ground truth is calculated using Eq. (6), and the results are illustrated in Fig. 2. It can be observed that the three most informative color channels are "*a*", "G", and "S". Also, to identify the most informative pair of color channels, all possible pairs of 10 color channels are considered. The latter experiment resulted in 45 states illustrated as a matrix in Fig. 3 where the least three informative 2-combinations of color channels are shown in red. The three most informative pairs of color channels can be considered as (*a*, B), (*a*, S), and (*a*, Gray) which are shown in green in Fig. 3. To identify the most informative 3-combinations of color channels, all possible 3-combinations of 10 color channels are considered. This experiment resulted in 120 states. According to this experiment, the three least informative 3-combinations of color channels are (R, H, V), (R, V, Gray), and (*l*, R, V). Moreover, the three most informative 3-combinations of color channels are (*a*, *l*, S), (*a*, G, S), and (*a*, S, Gray).

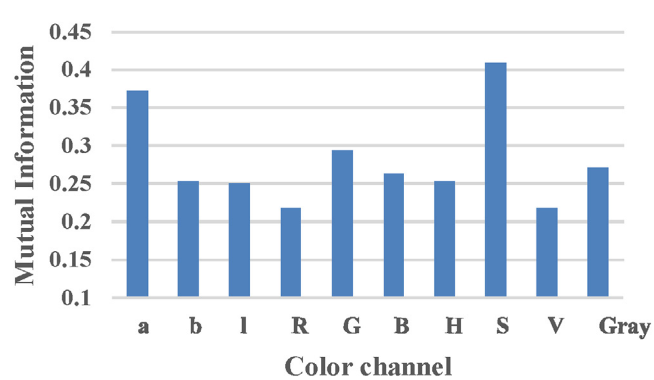

Fig. 2. Mutual information corresponding to single color channels.

	a	b	l	R	G	B	H	S	V	Gray
a	0.37	0.40	0.48	0.48	0.49	0.50	0.40	0.49	0.48	0.49
b	0.40	0.25	0.40	0.37	0.46	0.44	0.39	0.45	0.37	0.42
l	0.48	0.40	0.25	0.33	0.45	0.42	0.31	0.45	0.33	0.45
R	0.48	0.37	0.33	0.22	0.40	0.43	0.30	0.45	0.22	0.36
G	0.49	0.46	0.45	0.40	0.29	0.39	0.36	0.45	0.40	0.43
B	0.50	0.44	0.42	0.43	0.39	0.26	0.39	0.42	0.43	0.41
H	0.40	0.39	0.31	0.30	0.36	0.39	0.25	0.47	0.30	0.33
S	0.49	0.45	0.45	0.45	0.45	0.42	0.47	0.40	0.45	0.45
V	0.48	0.37	0.33	0.22	0.40	0.43	0.30	0.45	0.22	0.36
Gray	0.49	0.42	0.45	0.36	0.43	0.41	0.33	0.45	0.36	0.27

Fig. 3. Mutual information corresponding to 2-combinations of color channels.

3.2. Simplified MLP neural network

WCE devices suffer from limited computational capabilities and power resources and hence, simple MLP structures can be used more efficiently. As illustrated in Fig. 4, an MLP structure can be used for segmentation of the bleeding regions in WCE images. Three color channels containing the most information are extracted from a frame of WCE video. A patch is considered around each pixel, and extracted patches from the three color channels are aligned as inputs of the MLP. During the training phase, each patch is considered to be normal if and only if its central pixel is labeled as normal. Two neurons are considered as the output of the network, and a sotfmax activation function indicates the final class of the input patch. Considering resource limitations of the WCE device, no further feature extraction and preprocessing have been applied.

Quantization and pruning are effective methods to reduce the number of computational operations required by neural networks. Quantization reduces the bit-width of computational operations, and pruning reduces the number of computational operations.

For further simplification and preparation of the overall system for being embedded inside the capsule device, the network weights are quantized. In this paper, deterministic binary quantization of the weights is considered as follows:

$$W_q = \begin{cases} -1 & if\, W < 0 \\ & \square \\ 1 & if\, W \geq 0 \end{cases} \tag{7}$$

in which W and W_q are original and quantized network weights, respectively. Using the above quantization, all multiplications are removed and the amount of computation required for performing the segmentation is reduced significantly. The quantization algorithm is presented as pseudo code in Fig. 5. In Fig. 5, N is the number of epochs, L is the number of network layers, W_l is the weights matrix in layer l, W_l^q is its corresponding quantized weight matrix which is in form of binary values, C is the cost function, and η is the learning rate. As illustrated in the first line of Fig. 5, to achieve better performance, in the first epochs of training, variables are not quantized. After that, variables are quantized during training and loss is calculated using W_l^q. Finally, weights with real values are updated using gradients resulted from the back propagation step.

Train network parameters for a few epochs
For Epoch 1 to N
  For l=1 to L
    $W_l^q = Quantize\,(W_l)$
  END
  Forward and backward path using W_l^q
  For l=1 to L
    $W_l = W_l - \eta\,(\frac{\partial C}{W_l^q})$
  END
END

Fig. 5. Quantization algorithm for MLP.

3.3. CNN quantization and pruning

In many cases, problems arise during both pruning and quantization. While quantizing, bit length of variables reduces extensively which may result in problems in sections of the network with a

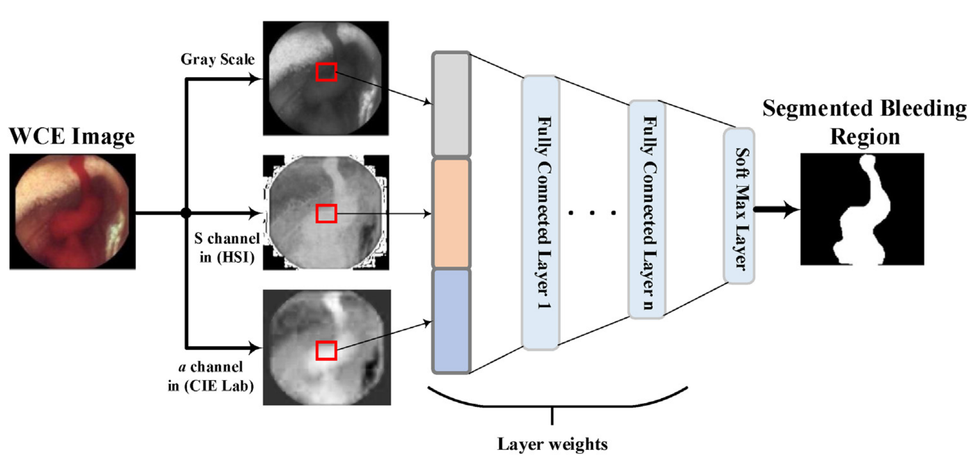

Fig. 4. Overview of the MLP architecture.

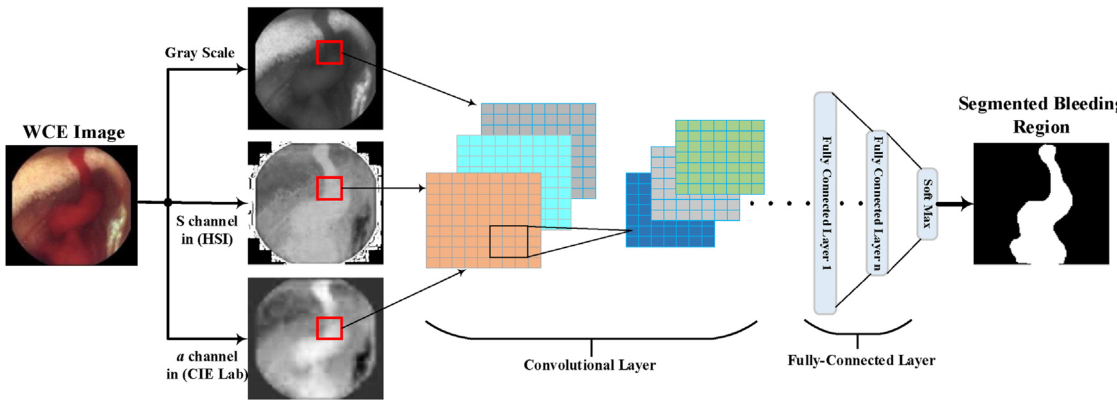

Fig. 6. Overview of the CNN architecture.

small number of parameters. For example in convolutional layers, a small number of coefficients are employed in each filter, and a large variation in filter coefficients due to the quantization leads to an undesirable effect on the network performance. Also, pruning has some problems due to the utilization of full precision operations. Although pruning methods remove redundant connections, pruned networks consume a lot of computational units when they are implemented on hardware because of using full precision parameters. Our proposed method addressed the above problems by employing the quantization and pruning of the CNN simultaneously. In proposed method, parameters of *fully connected layers* (FCLs) and *convolutional layers* (CLs) are quantized and pruned respectively.

To compensate for the accuracy loss due to the quantization and pruning, parameters are updated by retraining. In order to find better structural simplicity, pruning and quantization are performed simultaneously. In Fig. 6, a CNN architecture for segmentation of bleeding regions is illustrated. This structure is employed for simultaneous pruning and quantization. Image patches are selected from three color channels and used as inputs of the CLs which are followed by the FCLs. The weights of the FCLs are quantized and to avoid a severe drop of accuracy due to the simplification of CLs, their weights are pruned. Quantization of FCLs is based on using $\tilde{W} \in \{-1, 1\}$ instead of original network weight W. This quantization is performed during the training time using the algorithm in Fig. 5, and at the same time, CLs are pruned. Generally, the problem of simultaneous pruning and quantization of the CNN structure is illustrated in details as pseudo code in Fig. 7. Theoretical Justification of the proposed simplification method. In this pseudo code N is the number of epochs, L is the number of network layers and W_l is network weights matrix in layer l, W_l^q is its corresponding quantized weight matrix, $Mask_l$ is pruning mask for l, η is learning rate, C is the cost function and $\odot$ is the Hadamard production operation. At first, network parameters are trained in the conventional way and the pruning mask is obtained using Eq. (4). Convolutional layers are pruned based on the obtained pruning mask, and fully connected layers are binarized based on Eq. (7). The quantization and pruning steps cause an accuracy drop which may be alleviated during retraining.

By pruning of CLs, other remaining weights should be modified to compensate the effect of the removed weights. For a sample input $I \in \mathbb{R}^{C \times M \times N}$ and filter weight $w \in \mathbb{R}^{C \times M \times N}$, a single convolutional operation in CNN can be defined as Eq. (8).

$$y_i = \sum_{c=1}^{ch} \sum_{k_1=1}^{k_1=M} \sum_{k_2=1}^{k_2=N} I_{c,k_1,k_2} \times w_{c,k_1,k_2} \tag{8}$$

Train network parameters for a few epochs
Calculate pruning mask $(Mask_l)$
For Epoch 1 to N
  For l=1 to L
    IF W_l *is a conv filter*
      $W_l = W_l * Mask_l$
    ELSE W_l *is a FCL layer*
      $W_l^q = quantize\,(W_l)$
  END
  Forward and backward path using W_l *and* W_l^q *for convolutional filters and FCLs, respectively*
  For l=1 to L
    IF W_l *is a conv filter*
      $W_l = W_l - \eta\left(\left(\frac{\partial C}{W_l}\right) \odot Mask_l\right)$
    Else W_l *is a FCL layer*
      $W_l = W_l - \eta\left(\frac{\partial C}{W_l^q}\right)$
  END
END

Fig. 7. Simplification algorithm for CNN.

In which, y_i is the result of a single convolution (i^{th} convolution output), *ch*, is the number of input channels (filter channels), M is the filter width and N, is the filter height. For each layer of CNN, output feature maps are created using sliding window operations on input feature maps based on Eq. (8). For the sake of simplicity and without loss of generality, we assume the weights and corresponding input channels as the set $S = \{1, 2, \ldots, M \times N \times ch\}$. Also for the output of a single convolution, we assume that the input has similar size to set S. Therefore, Eq. (8) can be rewritten as Eq. (9).

$$y_i = \sum_{s=1}^{|S|} I_s \times w_s \tag{9}$$

In (9), $|.|$ represents the number of elements in S. In the process of pruning, some of un-necessary weights are removed and output $\tilde{y}_i$ is calculated using Eq. (10).

$$\tilde{y}_i = \sum_{s=1}^{|S'|} I_s \times w_s \tag{10}$$

In (10), it is assumed $S' \subset S$. Suppose that, n_o is the number of outputs, y_i is the i^{th} output of convolution and $I_{s,i}$ is s^{th} input corresponding to y_i. As a result, pruning is performed by finding a set $S' \subset S$ such that the Eq. (11) is satisfied.

$$\underset{|S'|}{arg\ min} \sum_{i=1}^{n_o} (y_i - \tilde{y}_i)^2 = \\ = \underset{|S'|}{arg\min} \sum_{i=1}^{n_o} \left(\sum_{s=1}^{|S|} I_{s,i} \times w_s - \sum_{s=1}^{|S'|} I_{s,i} \times w_s \right)^2 \tag{11}$$

y_i and $\tilde{y}_i$ are i^{th} results of convolution as Eq. (8) on set S and S' respectively. Defining pruned weights in S as the set $R = -S'$, the right side of (11) can be rewritten as:

$$\underset{|R|}{argmin} \sum_{i=1}^{n_o} \left(\sum_{s=1}^{|R|} I_s \times w_s \right)^2 \tag{12}$$

which is still NP hard, as R is unknown. If we know the size of, we can solve the optimization problem in Eq. (12) by a greedy search in which weights with smallest values are selected and added to R. In order to solve (12), suppose that the size of R is known. Although the resulting solution is suboptimal, solving Eq. (12) becomes possible. Therefore, for a specific number of removed filter weights, ($|R|$), we select the set R which leads to the smallest objective result in Eq. (12). Therefore, by selection of R a small variation is imposed to the network output and the following equation can be written as:

$$y_i = \sum_{s=1}^{|S|} I_s \times w_s = \sum_{s=1}^{|S'|} I_s \times w_s + \sum_{s=1}^{|R|} I_s \times w_s \tag{13}$$

Although the second term in right side of the Eq. (13) is selected to result in small variation in the network output, $\Delta x = \sum_{s=1}^{|R|} I_s \times w_s$ is imposed to the network and we have:

$$y_i = \sum_{s=1}^{|S|} I_s \times w_s = \sum_{s=1}^{|S'|} I_s \times w_s + \Delta x \tag{14}$$

To have the same results in the original and pruned networks, Δx should be compensated by modifying remained weights in the pruned network. In the proposed algorithm, Δx is compensated using retraining which is conducted after each pruning step. Since the weights with the smallest values are selected to meet Eq. (12), by selecting a proper pruning rate (in Eq. (4)), compensating the variations due to the pruning operations can be possible.

Quantization can be formulated in similar way of the pruning problem in which some variations are imposed to the network. In the proposed simplification method, quantization is performed for the FCLs. Neural computations for a single neural element (j^{th}) in a fully connected layer can be formulated as the following equation:

$$y_j = \sum_{n=1}^{n=N} O_n \times W_{n,j} \tag{15}$$

Where O_n is output of the n^{th} neural element and $W_{n,j}$ represents j^{th} weight from the n^{th} neural element. During the quantization process, $W_{n,j}$ is replaced with the quantized version $W^q_{n,j}$ and $\tilde{y}_j$ is resulted as the following equation:

$$y_j = \tilde{y}_j + \Delta x \tag{16}$$

In which $\tilde{y}_j$ is as follow:

$$\tilde{y}_j = \sum_{n=1}^{n=N} O_n \times W^q_{n,j} \tag{17}$$

And Δx is variation impose to the j^{th} neural element due to the quantization. In order to compensate Δx, weights of the quantized network are modified gradually. As illustrated in pseudo code in Fig. 7, learning of the FCL's weights is conducted after each quantization step. During training process, network gradients are calculated based on the quantized weights, but applied on the original ones. In this way, gradients are computed with the quantized weights but the original weights are used for the next quantization steps. Thus the original weights are modified and the effect of the quantization can be reduced. Also quantization effect is controlled via a small learning rate for better network training.

4. Experimental results

For the evaluation of the proposed method, experiments are performed using the TensorFlow framework. WCE bleeding images from publicly available databases are used [37–38]. Five bleeding images from [37], 50 bleeding and 728 normal images from [38] are used for training and testing. To study the performance of the proposed method, we use two metrics. The first metric is DICE score defined as:

$$DICE = \frac{2TP}{2TP + FP + FN} \tag{18}$$

where TP and TN are the numbers of pixels correctly classified as bleeding and non-bleeding, respectively. FP and FN are the numbers of pixels that are incorrectly classified as bleeding and non-bleeding, respectively. The second metric is *area under the curve* of ROC (AUC of ROC) which is used for evaluating the classification performance. Patch size is selected to be 9×9 experimentally. Experiments are performed mainly on two neural network structures including MLP and CNN as followings.

4.1. Imbalance data problem

Experiments on our WCE bleeding image dataset show that only 0.2% of the pixels belong to the bleeding class and the rest are non-bleedings. This is known as imbalance data problem. Imbalance data problem can be solved in the training phase by selecting the proper ratio of bleeding and non-bleeding patches. As most of the pixels are non-bleeding, training the neural network with the original dataset would result in a tendency toward the non-bleeding class and the network would not be able to classify the bleeding regions correctly. Therefore, for each network structure the ratio of bleeding and non-bleeding pixels is selected in a way that non-bleeding pixels do not overwhelm the bleeding pixels.

4.2. Simplified MLP configuration

Different MLP network configurations are tested, and the best architecture is selected. An MLP structure with three hidden layers consisting of 40, 20, and 8 neurons, respectively is applied as a simple structure for bleeding region segmentation. The sigmoid function is used as activation function, cross-entropy as loss function and Adam optimizer is used for optimization of the loss function. 500,000 patches are randomly selected in which the number of bleeding and the number of non-bleeding patches are equal.

Table 2
Segmentation performance in term of DICE score.

	Method	DICE	AUC of ROC
[8]	SVM	0.84	–
[9]	SVM	0.81	–
[10]	SVM	0.748	–
[14]	SVM	–	0.835
[7]	SVM	0.862	–
Our MLP (Quantized)	MLP-ANN	0.831	0.974
Our MLP (Full Precision)	MLP-ANN	0.861	0.983
Our CNN (Quantized)	CNN	0.846	0.978
CNN (Pruned -Quantized)	CNN	0.869	0.985
Our CNN (Full Precision)	CNN	0.890	0.984

4.3. Simplified CNN configuration

To evaluate the simplification method for CNN, a patch based CNN structure with 64 and 32 convolutional filters in each convolutional layer and fully connected layers of size 60 and 40 are used. The RELU activation function and pooling follow convolutional layers. Adam optimization method is used as an optimizer and cross-entropy as the loss function. The aforementioned configurations of the applied CNN are chosen by conducting numerous experiments using the same dataset as the MLP experiment to obtain the best results. Batch normalization and dropout are utilized for efficient training and dataset are balanced for better training. 600,000 patches are selected such that the bleeding patches are one-third of the normal patches.

4.4. Quantitative results

Different network structures and simplification methods are quantitatively compared in Table 2. Also, results of other related works on the segmentation of the bleeding regions in WCE images are reported in Table 2. Other works are based on different feature extraction methods and SVM for the classification. Since the number of normal pixels is much larger than the abnormal pixels, DICE score is used as the performance metric. For better comparison AUC of ROC metric is also included in Table 2. It can be observed that our full precision CNN have DICE score equal to 0.89. Selection of proper image color channels, using a balanced dataset, and configuration of an efficient CNN are the main advantages of the proposed full-precision CNN structure. As illustrated in Table 2, the simplified (quantized) MLP has about %3 drop in DICE score in comparison with the full precision MLP. Simplified MLP does not require any multiplications and has a simple structure in comparison with the CNN. If a more accurate model is demanded, it is possible to use a simplified CNN. From Table 2, it can be observed that quantizing all of the CNN parameters leads to about %4.5 drop in DICE score which can be reduced to about %2.2 with pruned-quantized CNN. Pruned-quantized CNN and full precision MLP have nearly similar results; however their structural complexities are very different. For better insight into the complexity of different network structures, structural complexities of the employed networks are analyzed in subsection IV.F. Also, the AUC illustrated in Table 2 shows that both pruned-quantized and full-precision CNNs have better segmentation results than the other methods. For further evaluation of the proposed method, ROC curves and AUC of ROC based on classification results of both MLP networks and all three versions of the CNN are presented in Figs. 8 and 9, respectively. MLP with original parameters (full-precision) and simplified MLP network have similar ROC curve in Fig. 8. Also in Fig. 9, the ROC curve of all CNN networks is similar to an AUC of about 0.98. Fig. 9 shows that a CNN has an appropriate segmentation capability. It is worth mentioning that MLP with a very simple structure has an acceptable AUC of ROC.

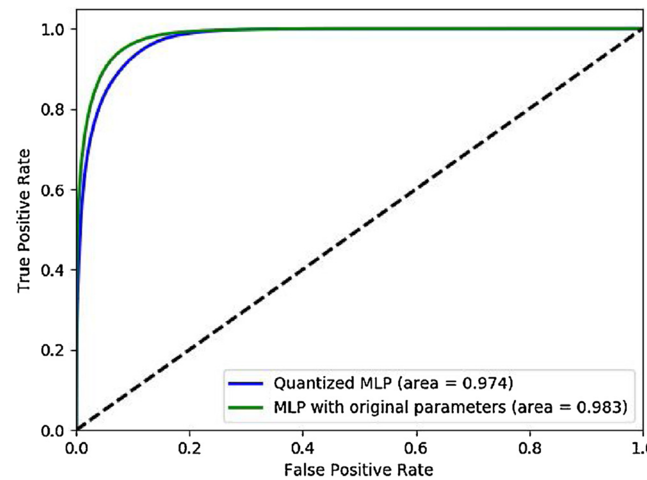

Fig. 8. ROC of MLP segmentation networks.

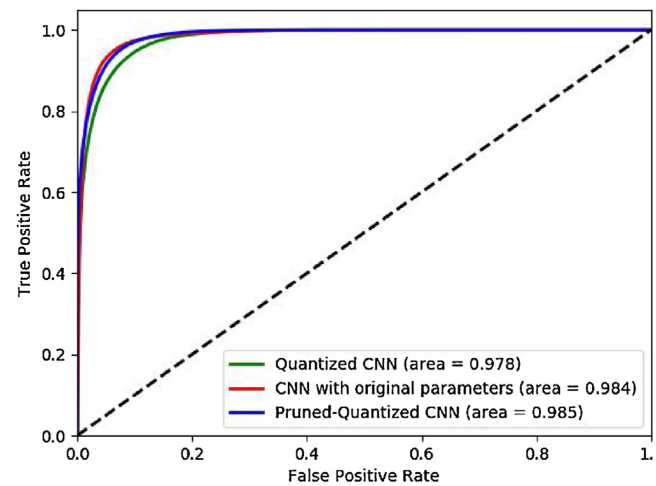

Fig. 9. ROC of CNN segmentation networks.

Table 3
Classification performance of the proposed method.

	Bleeding	Normal
Bleeding	55	0
Normal	37	691

For evaluating the performance of the proposed method in discriminating bleeding and normal frames, 728 normal images from the KID dataset as well as the previous bleeding images are used. We experimented the proposed CNN for detection of informative frames. Based on the proper segmentation results of the proposed network, after segmentation of the entire image, a simple thresholding method can be used to detect bleeding frames. To this aim, a threshold value of 200 is used to discriminate between bleeding and non-bleeding frames. Statistics regarding the performance of this thresholding method are illustrated in Table 3. From the 728 normal frames, only 37 frames are misclassified as bleeding frames and there is no misclassification in bleeding frames. From the total 783 frames with 55 informative frames, 691 frames are dropped and only 92 frames are transmitted. Dropping non-informative frames, can be regarded as an effective way to save energy resources in WCE device.

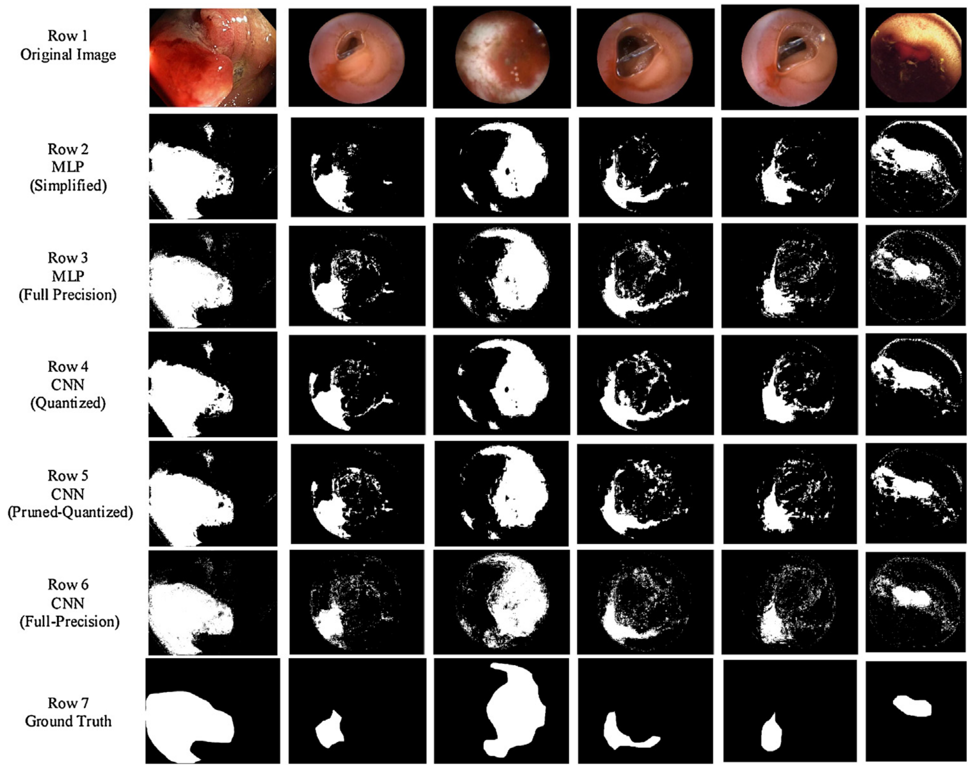

Fig. 10. Visual results of different networks for bleeding segmentation.

4.5. Qualitative results

In order to visually evaluate the proposed method, qualitative results of the full-precision and simplified MLP and the full-precision, quantized, and pruned-quantized CNN corresponding to six sample images of the utilized dataset are provided in Fig. 10. Segmentation results are in the form of a binary mask, and the ground truth of this mask is shown in row 7 of Fig. 10. Visual results of the simplified and full precision MLP are illustrated in the second and third rows of Fig. 10, respectively. Both MLP networks have acceptable visual results, and simplified MLP has no significant degradation of the visual quality in comparison with its full precision version. Also, the visual results of quantized, pruned-quantized, and full precision of CNN are illustrated in rows 4, 5 and 6 of Fig. 10, respectively. CNN with full precision parameters have the best visual results comparing with results of other networks. It can be observed in row 4 of Fig. 10, that CNN with only quantization has created some visible artifact due to the quantization error. As an example, there are some small regions in the ground truth resulted from the fully quantized CNN that are wrongly predicted as bleeding regions but the original CNN predicted them correctly. This error is alleviated in the visual result of the pruned-quantized network due to the efficient simplification method. Generally, two simplified network structures are considered including simplified MLP and pruned-quantized CNN with visual results illustrated in row 2 and 5 of Fig. 10. Pruned-quantized CNN network has better visual results, but the complexity of the simplified MLP network makes it suitable for implementation inside the capsule endoscopy device.

4.6. Complexity analysis

Suppose that the designed networks are going to be implemented on a device with limited hardware resource such as the capsule device. In devices such as WCE, implementing an algorithm which requires a small number of computational operations is very beneficial. For both of the proposed simplified networks, the computational complexity is considered as the number of multiplications and their bit-length and is reported in Tables 4 and 5, respectively. The number of parameters contained in each layer of both original and simplified MLP neural network is presented in Table 4 and also the number of parameters contained in each FCL and CL of all three versions of the utilized CNN is presented in Table 5. The number of network biases is negligible and has no significant effect on the total computational complexity; hence biases are ignored in Tables 4 and 5. In Table 4, for the MLP structure, all 32-bit multiplication operations are converted to 1-bit operations using quantization. 1-bit operations can be realized by addition without any multiplications that simplifies the computations needed by the network. Also, the types of layers in the utilized CNN are illustrated in Table 5. The number of parameters in full precision and simplified (pruned-quantized) CNN are compared. It is observed that the number of 32-bit length weights in full-precision CNN is reduced from 1728 to 979 in the first CL and from 18432 to 9163 in the second CL. Also, all of the weights in the FCLs are binarized.

Table 4
Summary of parameters in original and simplified MLP structure.

Layer	Type	Neurons	Original Weights	Simplified Weights
1	FC	40	9720	Quantized
2	FC	20	800	Quantized
3	FC	8	160	Quantized
4	FC	2	16	Quantized

Finally, all of the designed networks are compared with each other based on their complexity and DICE score in Table 6. In Table 6, the total number of employed parameters in terms of 32-bit, and 1-bit length operations are reported. Original CNN have 39,920 thirty-two-bit weights with DICE score equal to 0.890 which can be regarded as a complex network with an appropriate seg-

Table 5
Summary of parameters in original and simplified CNN structure.

Layer	Type	Maps and Neurons	Filter Size	Original Weights	Simplified Weights
1	Input	1M × 9 × 9	-	-	-
2	Convolution	64 M × 9 × 9	3 × 3	1728	979
3	Max Pooling	64 M × 5 × 5	2 × 2		-
4	Convolution	32 M × 5 × 5	3 × 3	18432	9163
5	Max Pooling	32 M × 3 × 3	2 × 2	-	-
6	FC	60	1 × 1	17280	Quantized
7	FC	40	1 × 1	2400	Quantized
8	FC	2	1 × 1	80	Quantized

Table 6
Number of parameters in two simplified networks.

Method	DICE score	# of parameters	
		32-bit	1-bit
Our original CNN	0.890	3̃9,920	0
Our original MLP	0.861	1̃0,696	0
Our simplified CNN	0.869	1̃0,142	1̃9,760
Our simplified MLP	0.831	0̃	1̃0,696

mentation performance. Original CNN parameters are reduced in simplified CNN (pruned-quantized CNN) which have 10,142 thirty-two-bit and 19,760 one-bit weights. Moreover, the simplified MLP is considered as a simple network with a lower segmentation performance in comparison with the CNN. This performance is improved in the original MLP network which have slightly more complex structure. Comparing the original MLP network and the simplified CNN, we can say that, in simplified CNN, more complex network is used with just 0.8 improvement in DICE score. Finally it can be said that CNN structures have better DICE score, while MLP networks have simple structures. Removing the heavy 32-bit operations and utilizing the binary operations reduce the network's computational complexity significantly and allows the inside-capsule implementation of the segmentation task.

5. Conclusion

In this paper, simplification methods for reducing the structural complexity of the neural networks for automatic segmentation of bleeding regions in WCE images were investigated. Two neural networks including MLP and CNN were applied for segmentation as different variations of neural networks' structure. The detection system was designed with respect to the limited resources of the WCE device. Hence, the MLP with quantized weights was utilized, and a new method based on pruning and quantization was employed for the CNN. In comparison with the original CNN, the pruned-quantized CNN needed almost 50% fewer parameters in CL, and also each of the 19760 weights in FCL was quantized. Furthermore, simulation results showed that the quantized MLP with 10696 weigh wts has comparable results with the simplified CNN. However, it needed much fewer parameters than the simplified CNN. Quantized neural networks without any multiplication could be considered as an automatic diagnostic approach inside the WCE device.

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